Data-Driven Box Office Analytics:

Movie Success & Rating Predictor

An End-to-End Supervised & Unsupervised Machine Learning Pipeline in Python

7th-Semester IT / **Data Science & ML/DL**

Python (Pandas, NumPy, Scikit-Learn, Matplotlib, Seaborn)

Thursday, **5 November 2026**

Problem Statement & Motivation

Building data-driven models to predict movie success and viewer ratings with machine learning and Python.

01
Predict
Regression models to forecast IMDb ratings (0–10 scale) and box office revenue.

02
Classify
Identify “Hit” vs “Flop” movies based on ROI > 2.5x.

03
Profile
Unsupervised clustering to discover movie archetypes and audience segments.

Dataset & Exploratory Data Analysis (EDA)

DATA OVERVIEW

TMDB and IMDb combined dataset with ~5,000 movies

KEY VARIABLES

Features

Production Budget

Runtime

Primary Genres

Director Popularity Score

Cast Star Power

Release Month

Targets

IMDb Rating (continuous)

Hit/Flop Status (binary)

EDA HIGHLIGHTS

Strong positive correlation between budget & box office revenue

Runtime and genre mix influence ratings (Drama/Biography > Horror/Action)

Distribution of ratings and revenue skewed; log transform applied in preprocessing

Workflow

01

Preprocessing & Cleaning

Prepare raw movie data for modeling.

02

Unsupervised Clustering

Discover movie archetypes from patterns.

03

Supervised Modeling

Predict IMDb ratings and classify hit vs. flop.

04

Evaluation & Interpretation

Measure performance and extract insights.

Experimental & Model

MODEL PERFORMANCE COMPARISON

MODEL	TASK	METRIC
Linear Regression	IMDb Rating (Regression)	R^2
		MAE
Decision Tree Classifier	Hit/Flop Classification	Accuracy
		F1-Score
Logistic Regression	Hit/Flop Classification	Accuracy

K-MEANS ARCHETYPES

0 **High-Budget Global Blockbusters**
High Revenue, Moderate Rating

1 **Low-Budget Critical Darlings / Indie**
Low Revenue, High Rating

2 **Commercial Mid-Budget Mainstream**

3 **Low-Budget Niche / Flops**

Algorithm Comparison & Justification

ALGORITHM	ROLE	WHY CHOSEN	LIMITATION
Linear Regression	Predict continuous IMDb rating score (0–10)	Fast, interpretable baseline; shows coefficient weight of each million dollars spent	Cannot capture complex non-linear interactions
Logistic Regression	Predict probability of Hit vs Flop	Clean probability percentages for risk assessment	Assumes a linear decision boundary
Decision Tree Classifier	Predict Hit vs. Flop using rule-based splits	Captures non-linear thresholds and transparent IF–THEN logic	Prone to overfitting without depth control
K-Means Clustering + PCA	Group unlabelled movies into 4 natural market archetypes	Discovers patterns without human bias; PCA reduces 15+ genre features to 2D	Unsupervised groups require interpretation

Python Tech Stack & Core Implementation

DATA PROCESSING	Pandas for data wrangling and feature extraction
MODEL BUILDING	Scikit-Learn for LinearRegression, LogisticRegression, DecisionTree, KMeans, and PCA
VISUALIZATION	Matplotlib & Seaborn for insightful data visualization

01

Load and preprocess movie datasets

↓

02

Train/test split (80/20) with Scikit-Learn pipeline

↓

03

Train models, predict, and evaluate performance

↓

04

Blend supervised predictions with unsupervised insights

Conclusion & Value

ACADEMIC IMPACT

Combing regression, classification, and clustering to turn data into meaningful insights in film analytics.

PRACTICAL APPLICATION

Enabling studios and streaming platforms to make objective, data-driven decisions and manage risk early.

FUTURE DIRECTIONS

Expanding with sentiment analysis of user reviews and an interactive Streamlit demo for real-time predictions.

Stop guessing if a movie is worth watching, let Python and Machine Learning do the math for you